Coding & Solution

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/suresh2006-del/EduGenie-AI.git
Programming Language(s)	Python 3.10+, HTML5, CSS3, JavaScript
Framework(s) Used	FastAPI, Jinja2 Templating, Uvicorn ASGI Server
Key Features Implemented	• Interactive Q&A & Simplification: REST endpoints (/qa, /explain) for academic doubt solving. • Quiz Generator: Dynamic practice test creation via /quiz. • Text Summarizer: Processing educational passages via /summarize. • Learning Recommendations: Structured learning path creation via /learn/recommendations. • Dual Model Execution: Google Gemini 1.5 Pro API integration along with offline fallback using LaMini-Flan-T5-783M model.
Pending / Incomplete Features	None. All core functional requirements, API endpoints, and UI views have been fully implemented.

Setup / Run Instructions	1. Clone repository: git clone https://github.com/suresh2006-del/EduGenie-AI.git 2. Install dependencies: pip install -r requirements.txt. 3. Set API key: export GEMINI_API_KEY="AQ.Ab8RN6IzBASX9Lki8kIevSIu-W_1_53R4Ju99c_uHTc9mhXwjg". 4. Start server: uvicorn main:app --reload. 5. Access application at (http://127.0.0.1:8000) in browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The solution is fully functional and optimized to run efficiently on standard hardware setups (including Apple Silicon / Mac M1 and dual-core CPUs) without requiring external GPU hardware. Robust exception handling ensures graceful error recovery during API network disconnects by falling back to local model pipelines.